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DANIEL HERIBERTO **PALENCIA ARREOLA**

CURRICULUM VITAE

CONTENTS

PROFESSIONAL SUMMARY

Systems-oriented Engineering Leader & Venture Founder with a 12+ year track record of scaling global technical organizations from 0 to 1 across Asia, Australia, and the Americas. Expert in transforming legacy technical debt into high-availability enterprise stacks and bridging the gap between **MIT-tier research and commercial viability**. A strategic architect of **socially responsible technology**, I have led the development of complex platforms for **renewable energy, carbon tracking, and urban mobility**. Proven success in securing **$250k+ in non-dilutive innovation funding** and building high-retention, cross-functional cultures that align engineering excellence with meaningful environmental and social impact.

PROFESSIONAL EXPERIENCE

Pulse Clinic — Singapore & Bangkok, Thailand
Chief Technology Officer (CTO) — 2025 – Now

```
$ {C#, TypeScript, Docker, Agile, CD/CI, Terraform, GitHub Actions, Jira}
```

- **Engineering Transformation:** Re-engineered a decade of legacy technical debt by implementing CI/CD, Agile methodologies, and Clean Code principles.
- **Organizational Growth:** Built a cross-functional Agile team (UX/Product/Eng) by vetting hundreds of candidates to transform a developer-only silo.
- **System Reliability:** Improved platform uptime by over 90%, reducing critical outages from 3x weekly to <1x monthly.

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- **Regional Leadership:** Strategized technology for 200+ employees serving tens of thousands of customers across 6 international markets (TH, MY, SG, PH, HK).
- **Digital Transformation:** Architected and executed migrating from a legacy in-house Clinic Management System to an enterprise solution with zero data incidents.

Flo Energy — Singapore & Melbourne, Australia
Tech Lead -> Engineering Manager — 2021 – 2025

```
$ {Kotlin, TypeScript, GraphQL, Kubernetes, Docker, Agile, CD/CI, Terraform, Bitbucket,
    Jira, IoT}
```

- **Organizational Scaling:** Built 4 global engineering teams from zero, scaling from 4 to 20+ members in 1.5 years with established onboarding and innovation culture.
- **Product Expansion:** Oversaw 0-to-1 product development and expansion into international markets, including Renewable Energy Certificate implementations.
- **Strategic Integration:** Led high-stakes technical integrations with Singapore's Energy Market Company and drove evidence-based corporate decision-making.
- **Operational Excellence:** Hardened infrastructure by modernizing CI/CD pipelines, automated testing, and global privacy/security policies.
- **Technical Strategy:** Directed quantitative research and corporate deliberation to steer high-level technology improvements and IoT architecture.

Singapore - MIT Alliance for Research and Technology (SMART) — Singapore
Tech Lead Software Engineer — 2020 – 2021

```
$ {Node.js, Angular, Express, TypeScript, WSL, Linux, Ubuntu, ECharts, Open Layers, Python}
```

- **System Modernization:** Architected and engineered a system-wide UI overhaul for SimMobility's agent-based simulation platform, migrating legacy components to a high-performance Angular framework.

Michoacan University of Saint Nicholas of Hidalgo (UMSNH) — Morelia, Mexico
Lecturer - Urban Data — 2020 – 2021

```
$ {ArcGIS, QGIS, CARTO, Open Data}
```

- **Curriculum Design:** Developed the original syllabus and technical materials for a graduate-level Urban Data course from the ground up.
- **Geospatial Instruction:** Taught Open Data philosophy and practical urban planning applications using QGIS and CARTO mapping suites.

Massachusetts Institute of Technology (MIT) + SMART — Boston, USA & Singapore
Project Lead Software Engineer — 2014 – 2020

```
$ {Android, Java, JavaScript, Mapbox API, PostgreSQL, REST, Spring, Jersey, Google Maps
    API, PostGIS, R, R Studio}
```

- **Team Leadership:** Led a team of 9 in the R&D of Flocktracker, a specialized platform for high-precision, georeferenced field surveys.
- **Full-Stack Architecture:** Architected and programmed the end-to-end cloud stack, including the RESTful backend and the Android client-side UX.
- **Engineered Resilience:** Developed a custom asynchronous system for geospatial data collection and route tracing, optimized for unreliable offline environments.
- **Global Scale:** Directed community outreach and negotiated software deployments in 30+ locations across 4 continents.
- **Technology Commercialization:** Successfully translated research into a commercial venture by launching Fuente to take Flocktracker to market.

Kota Kita Surakarta, Indonesia
Technologist 2014 – 2014

```
$ {Mapbox, Database Design, Web Map Design}
```

- **Civic Technology:** Designed and developed a participatory process database and technical infrastructure for urban engagement.
- **Geospatial Visualization:** Engineered front-end web map designs to visualize community-driven data.

Urban Launchpad + New Market Goods San Francisco, USA & Dhaka, Bangladesh
Technologist 2014 – 2014

```
$ {Android, Mapbox, Web Dev, Kickstarter}
```

- **Technical Strategy:** Acted as strategy consultant and lead developer for a Bay Area social startup, overseeing front-end web and Android programming.
- **Crowdsourcing launch & Data Transparency:** Designed a transparency database and produced all technical materials for a successful Kickstarter crowdfunding campaign.

MIT + Urban Travel Logistics Mexico City, Mexico
Software Engineer 2013 – 2013

```
$ {Android, System Research}
```

- **Mobile Research Systems:** Developed and modified specialized Android applications to conduct system research on urban transport logistics.
- **Operations & Scholarship:** Managed large-scale data collection operations and contributed to academic scholarship regarding software research methodologies.

Engineering Institute, National Autonomous University of Mexico Mexico City, Mexico
Research Assistant 2011 – 2014

```
$ {FORTRAN, AutoLISP, AutoCAD, Finite Element Modeling, Discrete Element Modeling}
```

- **Geotechnical Modeling:** Developed 3D CAD tools for finite and discrete element modeling of soil dams and shafts under seismic loads using FORTRAN and AutoLISP.

ENTREPRENEURSHIP

Fuente Tech Singapore
Founding Chief Executive Officer (CEO), Tech Lead 2020 – 2026

```
$ {Android, Java, Gradle, GitHub Actions, Docker, AWS, Mapbox, PostgreSQL, Spring, Jersey}
```

- **Capital Acquisition:** Raised $200,000+ USD in non-dilutive funding from the National Science Foundation (NSF), MIT, and SMART.
- **IP Strategy:** Negotiated complex technology licensing agreements for the Flocktracker platform with MIT and the National University of Singapore (NUS).
- **System Hardening:** Architected and implemented thousands of unit and integration tests to ensure high availability for enterprise-grade deployments.
- **Cloud Infrastructure:** Orchestrated the migration of the Flocktracker stack to AWS to transition the platform from research to commercial scale.

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M+B — Singapore
Founding Chief Technology Officer (CTO) — 2020 – 2021

```
$ {WordPress, WPML, Stripe, WooCommerce}
```

- **Platform Engineering:** Designed and implemented a multilingual publication platform using WordPress and WPML.
- **E-commerce Automation:** Built a custom web store to fully automate the sales lifecycle, from end-user purchase through to fulfillment and accounting.

Joro (now Commons Earth) — Boston & Oakland, USA
Founding Software Advisor — 2017 – 2020

```
$ {Node.js, React, React Native, MongoDB}
```

- **Technical Foundation:** Designed database and full-stack architecture for a React Native carbon-footprinting platform, incorporating AI-driven UX features.
- **Strategic Advisory:** Guided software architecture, documentation standards, and engineering hiring to scale the technical organization.

EDUCATION

Massachusetts Institute of Technology (MIT) — Boston, USA
Master of City Planning — 2016–2019
Urban data, mobility and sustainability

Autonomous Metropolitan University (UAM) — Mexico City, Mexico
Studies in Social Anthropology — 2011–2013
Urban conflict

National Autonomous University of Mexico (UNAM) — Mexico City, Mexico
Civil Engineer — 2009–2012
Soil and seismic engineering

VENTURE & INNOVATION FELLOWSHIPS

Smart – Innovation Centre | **Innovation Grant** | **$100,000 SGD** — Singapore, 2019–2020

Selective go to market research commercialization grant.

National Science Foundation | **Innovation Corps** | **$50,000 USD** — Detroit, USA, 2018–2020

Conducted 150+ primary market research interviews in the US and Mexico.

MIT – Deshpande Center | **Innovation Grant** | **$100,000 USD** — Boston, USA, 2016–2018

Selective grant to support the commercialization of Flocktracker.

PUBLICATIONS

Yun, HY, Zegras, PC, Palencia, DH	**"Digitalizing Walkability": Comparing Smartphone-Based and Web-Based Approaches to Measuring Neighborhood Walkability in Singapore** *Journal of Urban Technology*	Singapore 2019
Palencia, DH	**Arguments for and Field Experiments in Democratizing Digital Data Collection – The Case of Flocktracker** *Massachusetts Institute of Technology*	Boston, USA 2019
Sit, M, Alvarez, R, Palencia, DH, Duarte, F, Ahn, C, Zegras, C, Biderman, A, Ratti, C	**Citivan: Crowdsourcing User Experience Feedback via Mobile Phone Survey Platforms to Monitor Service Quality Levels in Public Transportation Systems** *Future Urban Mobility Symposium*	Singapore 2017
Zegras, PC, Butts, K, Cadena, A, Palencia, DH	**Spatiotemporal Dynamics in Public Transport Personal Security Perceptions: Digital Evidence from Mexico City's Periphery, in ICT for Transport: Opportunities and Threats (P. Rietveld, M. Givoni, N. Thomopoulos, Eds.)** *Edward Elgar Publishing*	USA 2015
Yun, HY, Zegras, PC, Palencia, DH, Chong, K	**"Digitalizing Walkability": Examining the Value of Using a Smartphone-Based Approach to Measure Neighborhood Walkability in Singapore** *H Future Urban Mobility Symposium*	Singapore 2015
Palencia, DH, Cadena, A, Butts, K, Zegras, PC	**Flocktracker: Example Applications in Urban Environments (Poster)** *Future Urban Mobility Symposium*	Singapore 2015
Palencia, DH, Cadena, A, Butts, K, Zegras, PC	**Flocktracker: Technical Components (Poster)** *Future Urban Mobility Symposium*	Singapore 2015
Palencia, DH	**Método Analítico para el Cálculo de Presiones Laterales y Elementos Mecánicos en Lumbreras Sometidas a Carga Estática y Sísmica** *Universidad Nacional Autónoma de México*	Mexico City, México 2013
Palencia, DH, Méndez BC, Botero E, Romo MP	**Analytical Method to Calculate Lateral Pressures on-, and Mechanic Elements of- Shaft Linings** *14o Conferencia Panamericana de Mecánica de Suelos e Ingeniería Geotécnica*	Toronto, Canada 2012
Palencia, DH, Botero E, Romo MP, Oropeza, C, Ponce, S	**Método para la Generación de Mallas Tridimensionales a Partir de Información Topográfica** *XXVI Reunión Nacional de Mecánica de Suelos e Ingeniería Geotécnica*	Cancún, México 2012
Méndez BC, Palencia DH, Botero E, Romo MP	**Numerical Computation of Shaft Lining Pressures in Granular Soil** *The 13th International Conference of International Association for Computer Methods and Advances in Geomechanics (IACMAG) 9–11, Paper id 1149*	Melbourne, Australia 2011

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Palencia, DH, Méndez BC, Botero E, Romo MP	**Método Analítico para el Cálculo de Presiones Laterales y Elementos Mecánicos en Recubrimientos Lumbreras** *XXV Reunión Nacional de Mecánica de Suelos e Ingeniería Geotécnica*	Acapulco, México 2010

OTHER SKILLS

Machine Learning / AI / Data Science

Algorithms:

```
$ {CART, K-Means Clustering, Linear Regression, Logistic Regression,
   Random Forests, Text Analytics}
```

Tools:

```
$ {Python, R, R Studio, D3, GGPlot2}
```